Community Structures in Financial Markets

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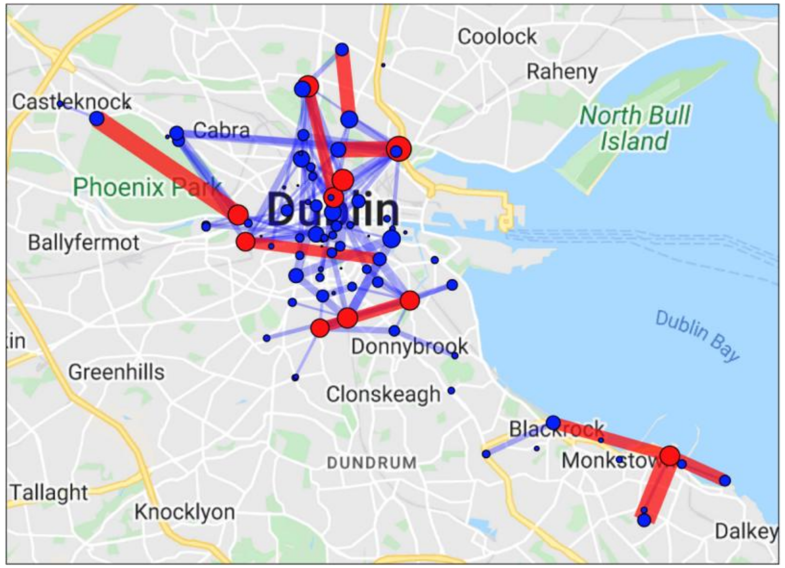
Network (Graph)

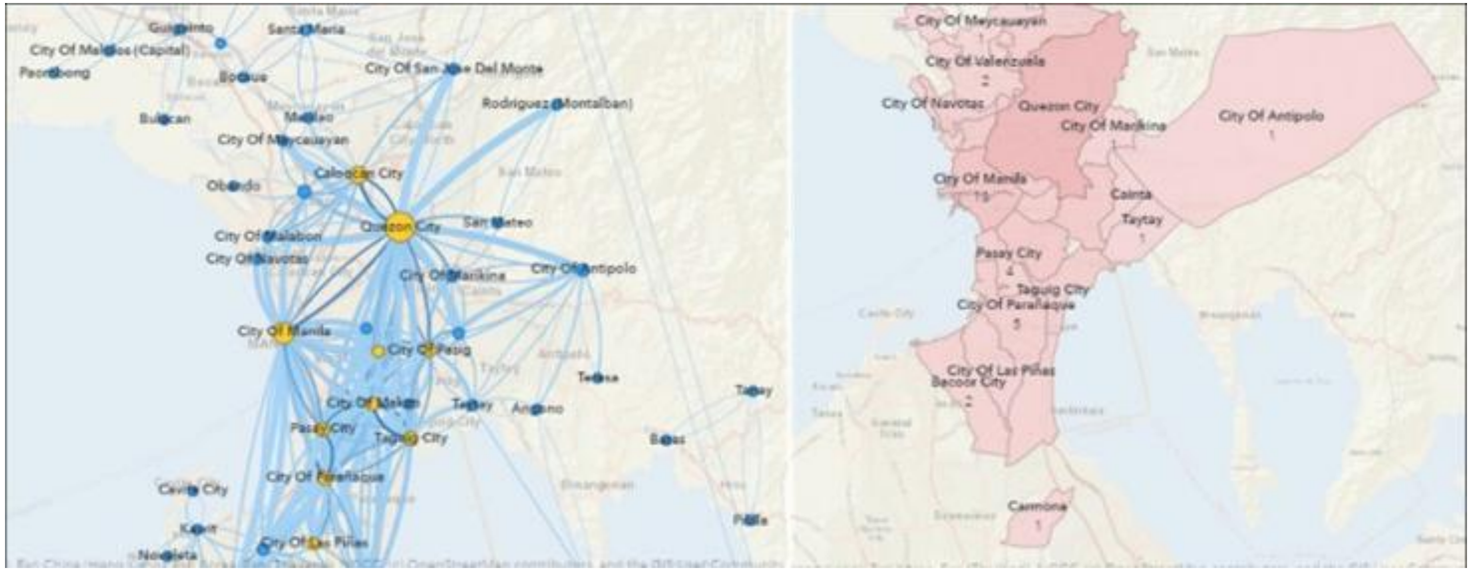
Definition

A **network (graph)** consists of **nodes**, **edges** and their corresponding **weights** where each node represents an object of a system and an edge connects 2 nodes together with a corresponding weight measuring their relationship.

Useful?



Shared bike usage



Virus Spread Trends

Community

Definition

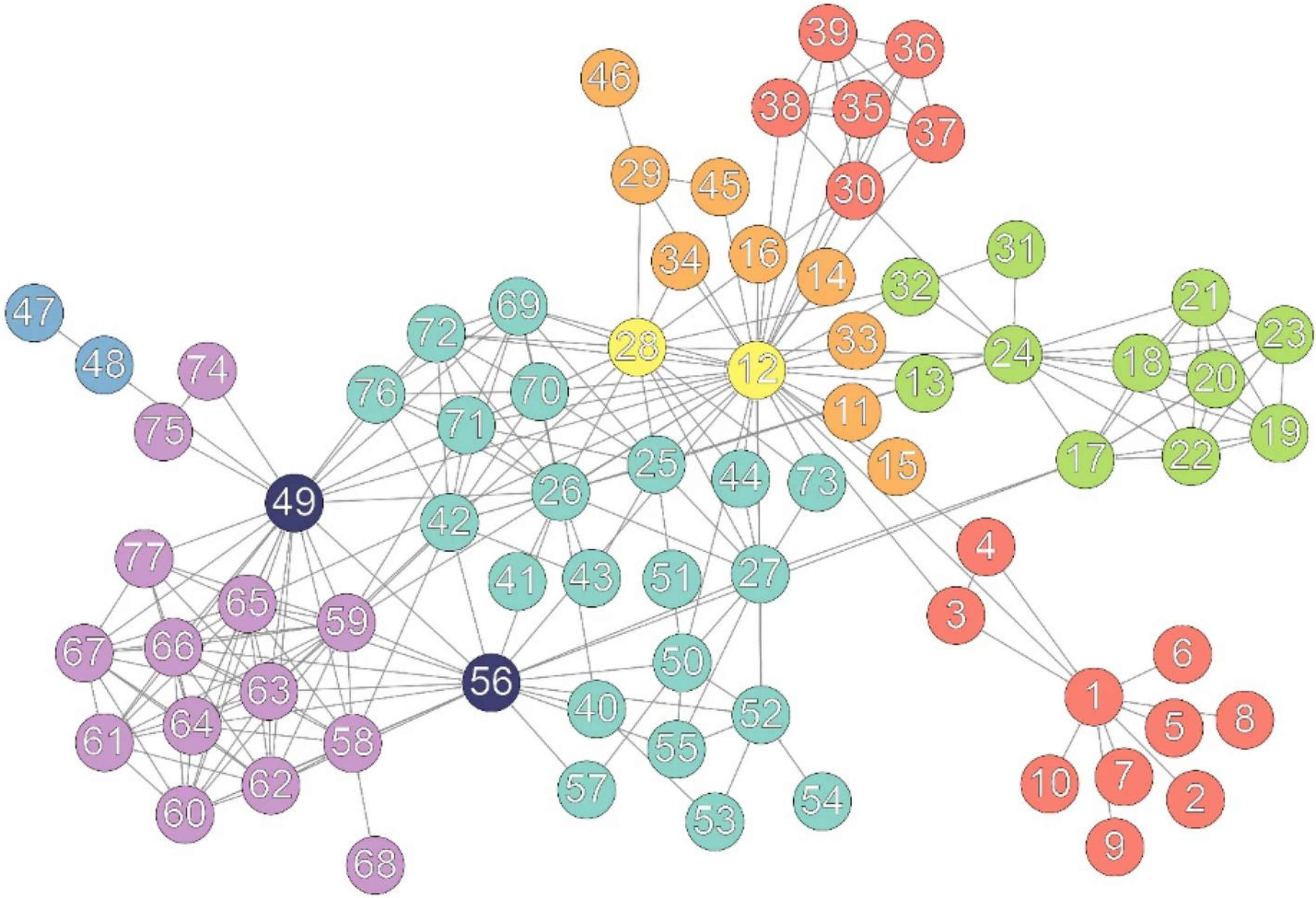
A group of nodes in a network that are more densely connected to each other than to the rest of the network

Why community detection?

To facilitate a more comprehensive analysis of network characteristics

Methods

Louvain, Girvan-Newman, K-means, Clique Percolation Method, etc



Note!

Looking at the community and network structure alone may not provide enough insight. To gain a deeper understanding, it's important to incorporate additional indicators such as trading volume, ranking, and macroeconomic factors, as well as graph-based metrics to support and enrich the community analysis.

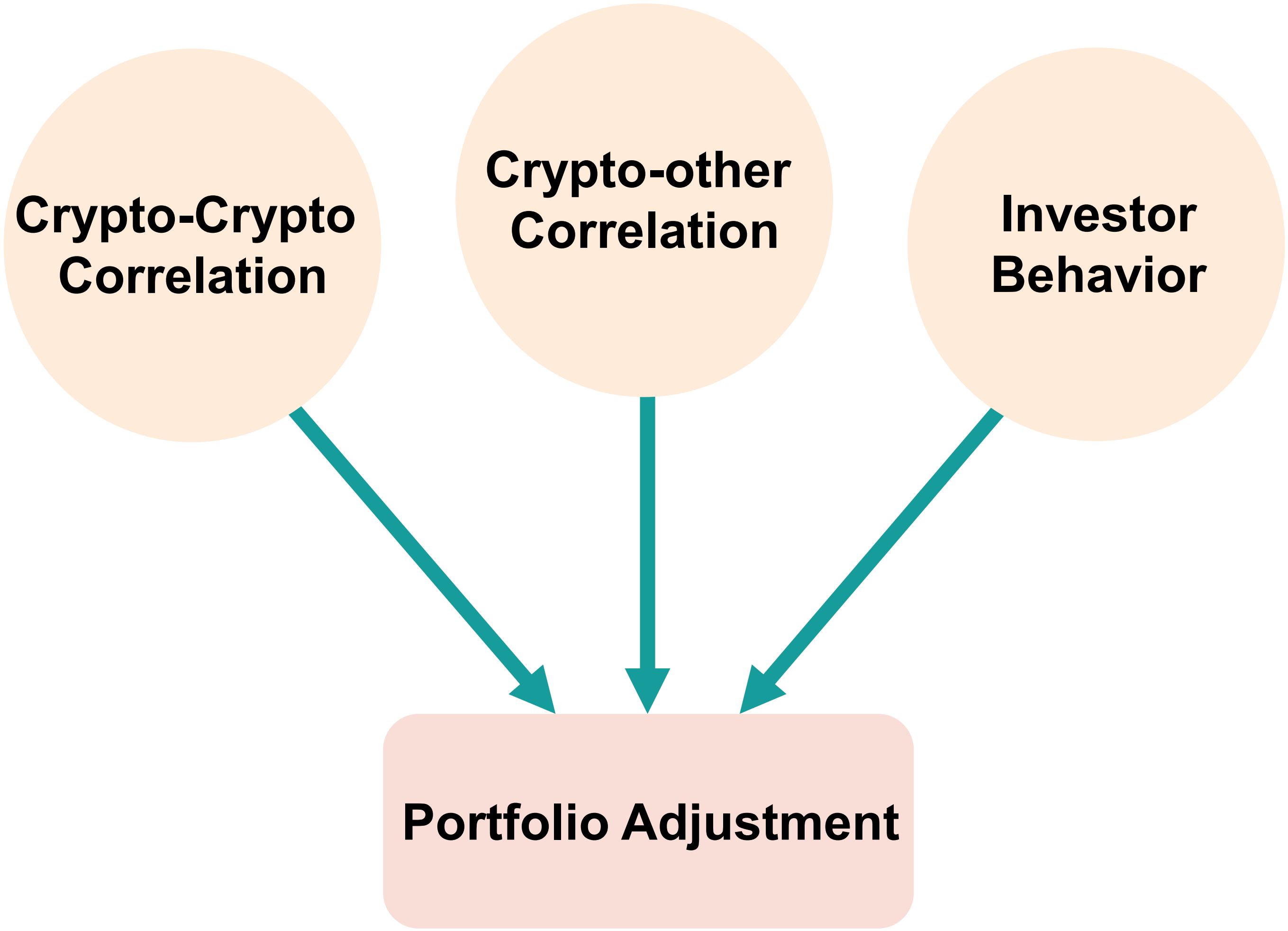
Network and Community Structures in Financial Markets

Investment Vehicles Covered

Stock: company-based, representing the growth and potential of a company. For example, AAPL, MSFT, NVDA.

Crypto: investor behavior-based, sensitive to investors' reactions. For example, BTC, ETH, USDT.

ETF: A basket of stocks, representing a general growth of the stocks. For example, SP500, QQQ, SOXX.

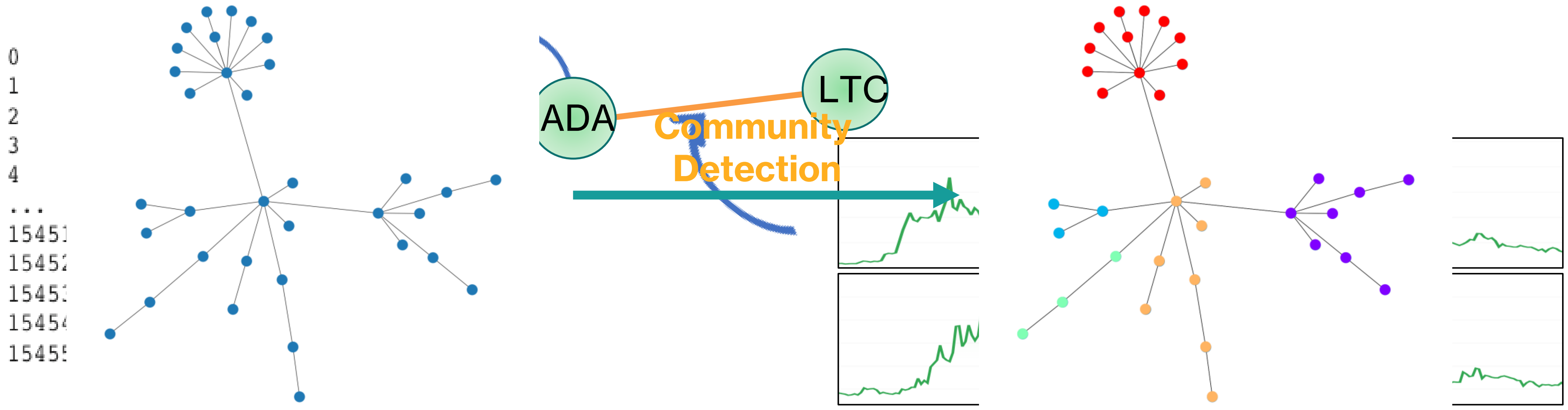


Community Structures in the Cryptocurrency Market

Correlation Network and Community Detection

In financial markets, each node is an asset and an edge between two nodes measures their **correlation** represented by a weight.

This **correlation** represents the **co-movement** of the assets.



Community Structures in the Cryptocurrency Market

Targets

1. How does the structure of the return-based correlation graph evolve over time? In particular, does it differ between stable and turbulent periods?

2. If so, could these changes be linked to shifts in investors' strategies? In other words, does the investment behavior of investors influence the correlation?

Dataset

Tick-by-tick dataset consisting of 34 cryptocurrencies with a hybrid of high and low rankings (accessed in April 2021, Coinmarketcap.com)

- Highest ranking: Bitcoin (1)
- Lowest ranking: Funtoken (196)

Period is from 13/02/2019 to 06/04/2021

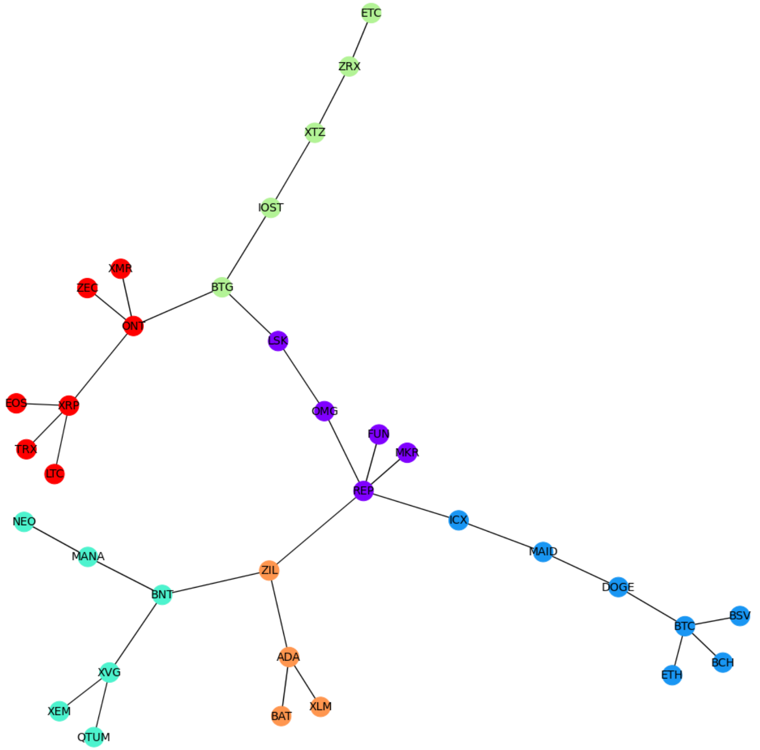
Timescale: 30 mins

Timeline Division

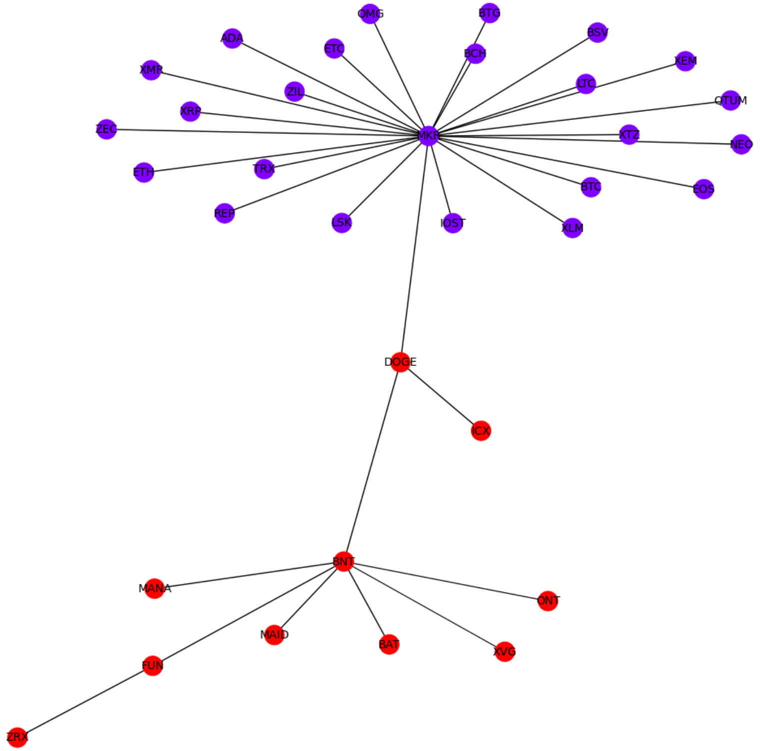
Time Window	Stage	Time Span	# Days
1	Normal time	13 February 2019–31 December 2019	322 days
2	Downturn time	1 January 2020–30 June 2020	182 days
3	Recovery time	1 July 2020–6 April 2021	280 days

Community Structures in the Cryptocurrency Market

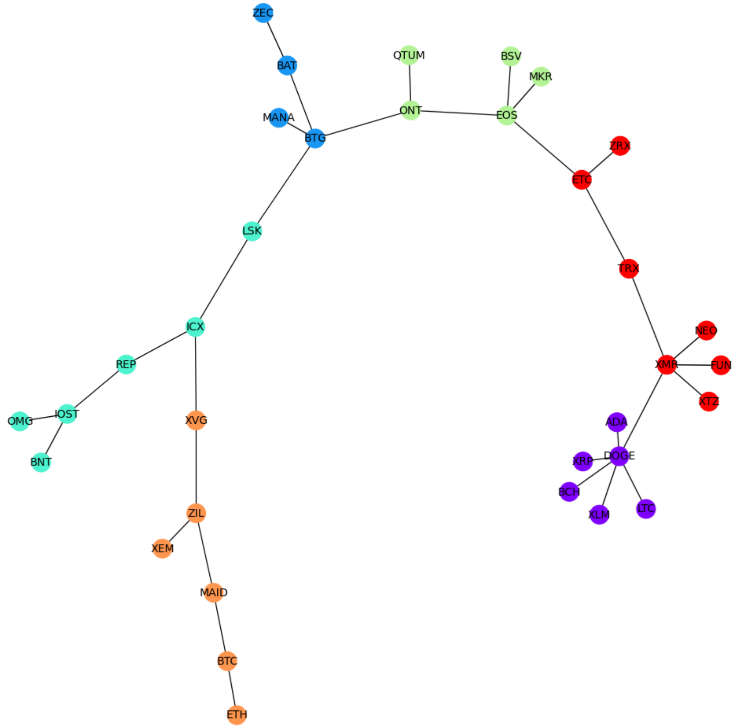
Results – Network Structures



Time window 1
13/02/2019-31/12/2019
Normal time



Time window 2
01/01/2020-30/06/2020
Global downturn



Time window 3
01/07/2020-06/04/2021
Recovery time

Community Structures in the Cryptocurrency Market

Results - Graph-based Metrics

Table 1.1: The growth of network structures over time measured by Betweenness Centrality and Degree Assortativity.

Metrics	Time Window 1	Time Window 2	Time Window 3
Betweenness centrality	0.15	0.05	0.16
Degree Assortativity	-0.49	-0.72	-0.51

Table 1.2: Similarity in network structures between different phases of the cryptocurrency market measured by three metrics.

Metrics \ Time Window	1 vs. 2	1 vs. 3	2 vs. 3
Degree centrality	0.5	0.09	0.42
Eigenvalue method	844.45	4.59	759.16
v-measure	0.04	0.32	0.02

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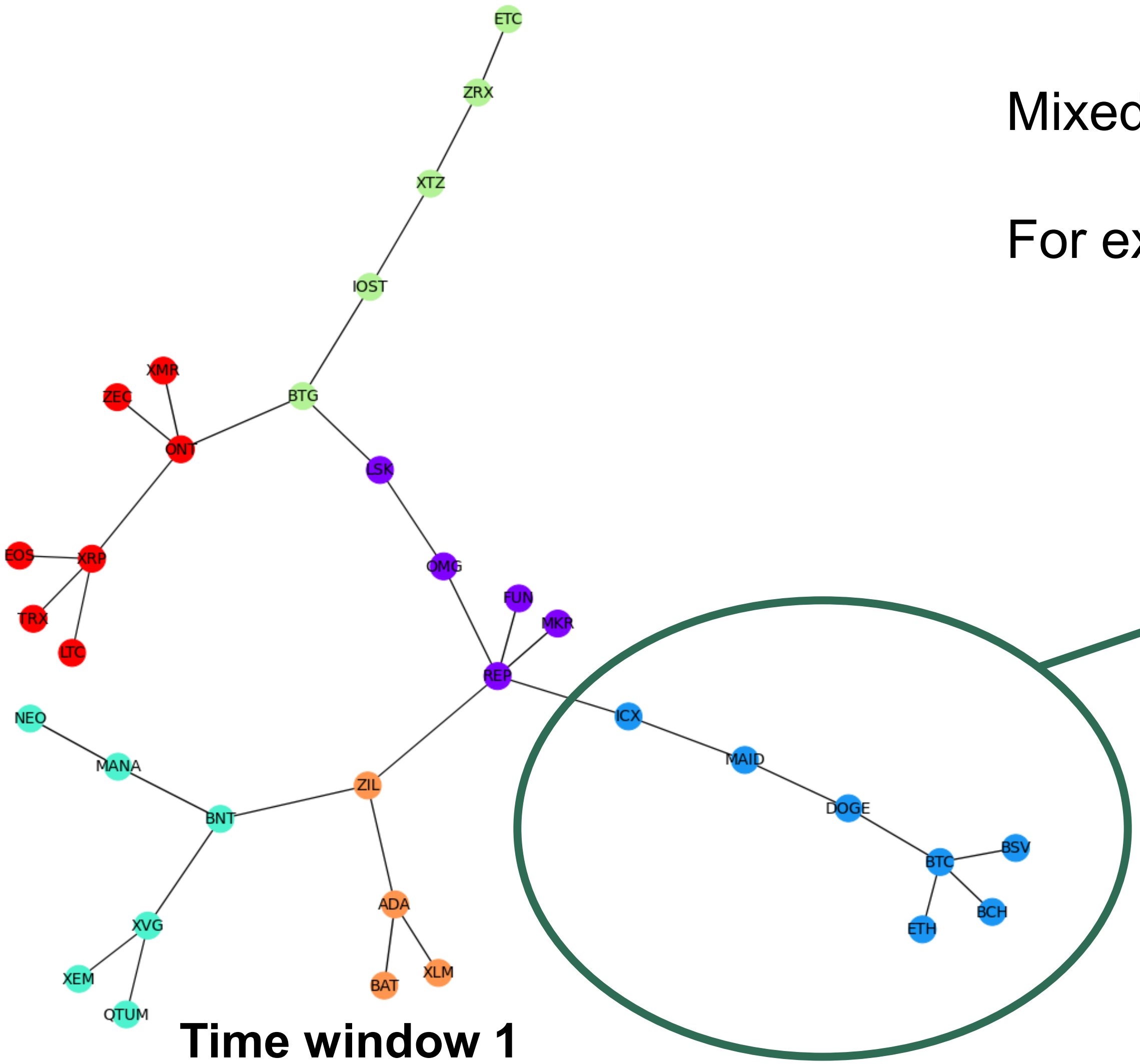
Results – Ranking Distribution



Time window 1
13/02/2019-31/12/2019
Normal time

Mixed high and low-ranking assets in each community

For example,

BTC: 1
ETH: 2
BCH: 5
BSV: 9
DOGE : 34
MAID: 84
ICX: 130

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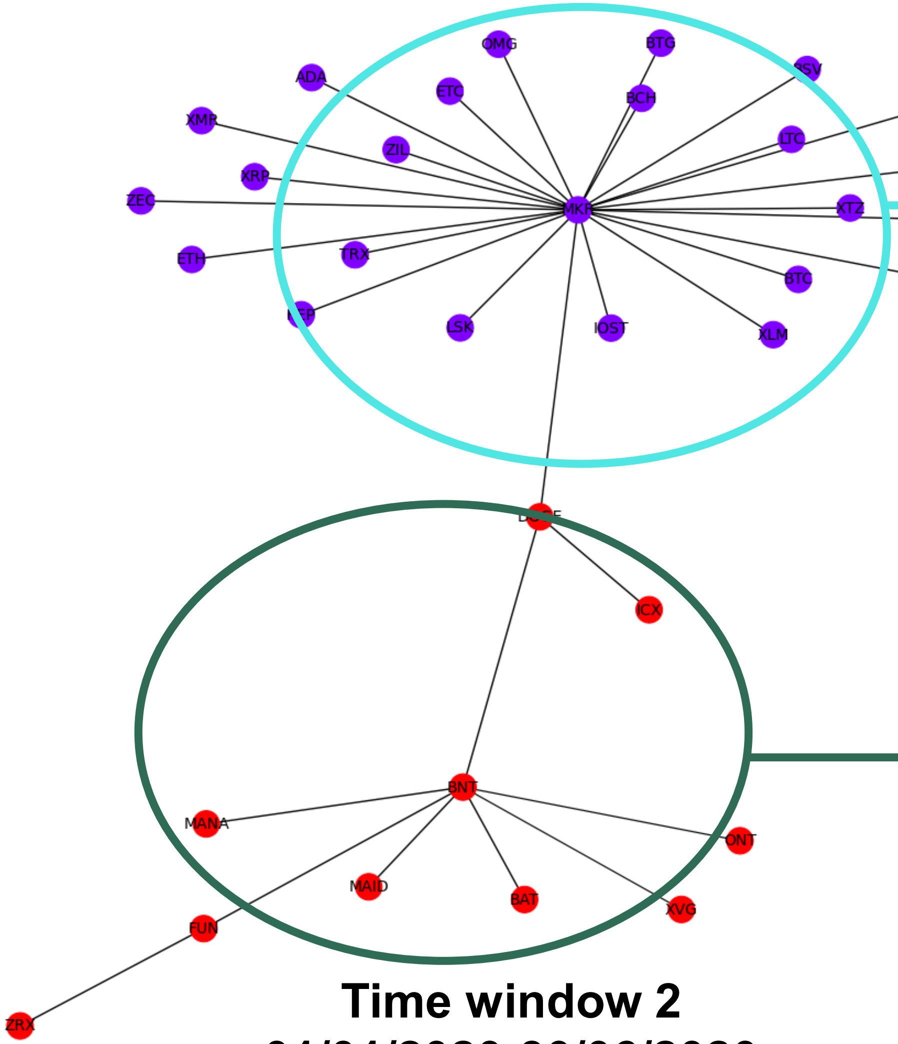
Results – Ranking Distribution



Rankings from 1 to 30

Rankings from 60 to 196

High-ranking assets belong to one group and low-ranking assets belong to the other.

Time window 2
01/01/2020-30/06/2020
Global downturn

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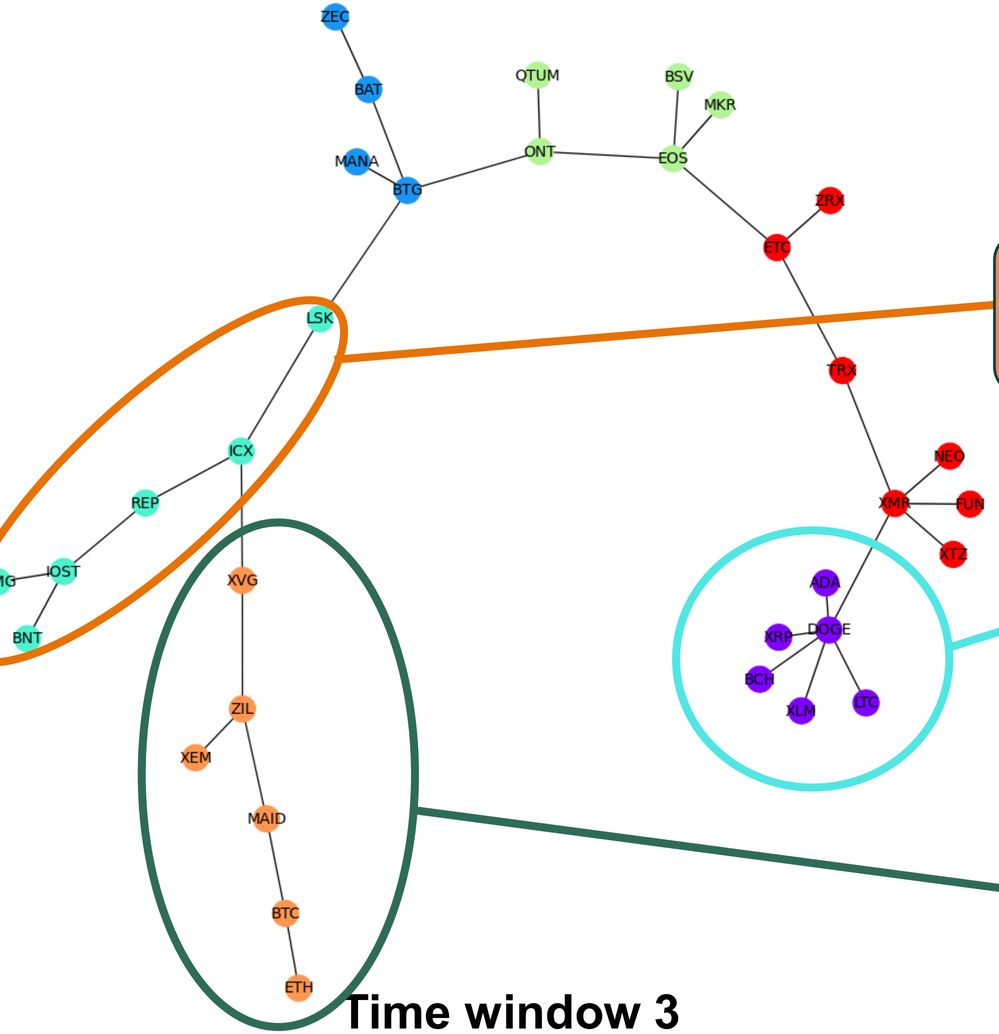
Results – Ranking Distribution



More diversified, exhibiting behavioral patterns from both the first and second time windows.

Rankings from 68 to 140

Rankings from 1 to 20

BTC: 1
ETH: 2
XVG: 38
ZIL: 48
MAID: 109
XEM: 138

Time window 3
01/07/2020-06/04/2021
Recovery time

Community Structures in the Cryptocurrency Market

Conclusion 1

The behaviour of investors in making investment decisions can be considered a potential explanation for the time-varying graph structures:

- During normal times: contrarian behavior/risk-taking.
- During the pandemic: risk-aversion.
- During recovery time: risk-taking and risk aversion.

Community Structures in the Cryptocurrency Market

Can we investigate deeper?

Time Window	Stage	Time Span	# Days
1	Normal time	13 February 2019–31 December 2019	322 days
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Instead, using degree assortativity, betweenness centrality and closeness centrality to assess the structure of a graph.

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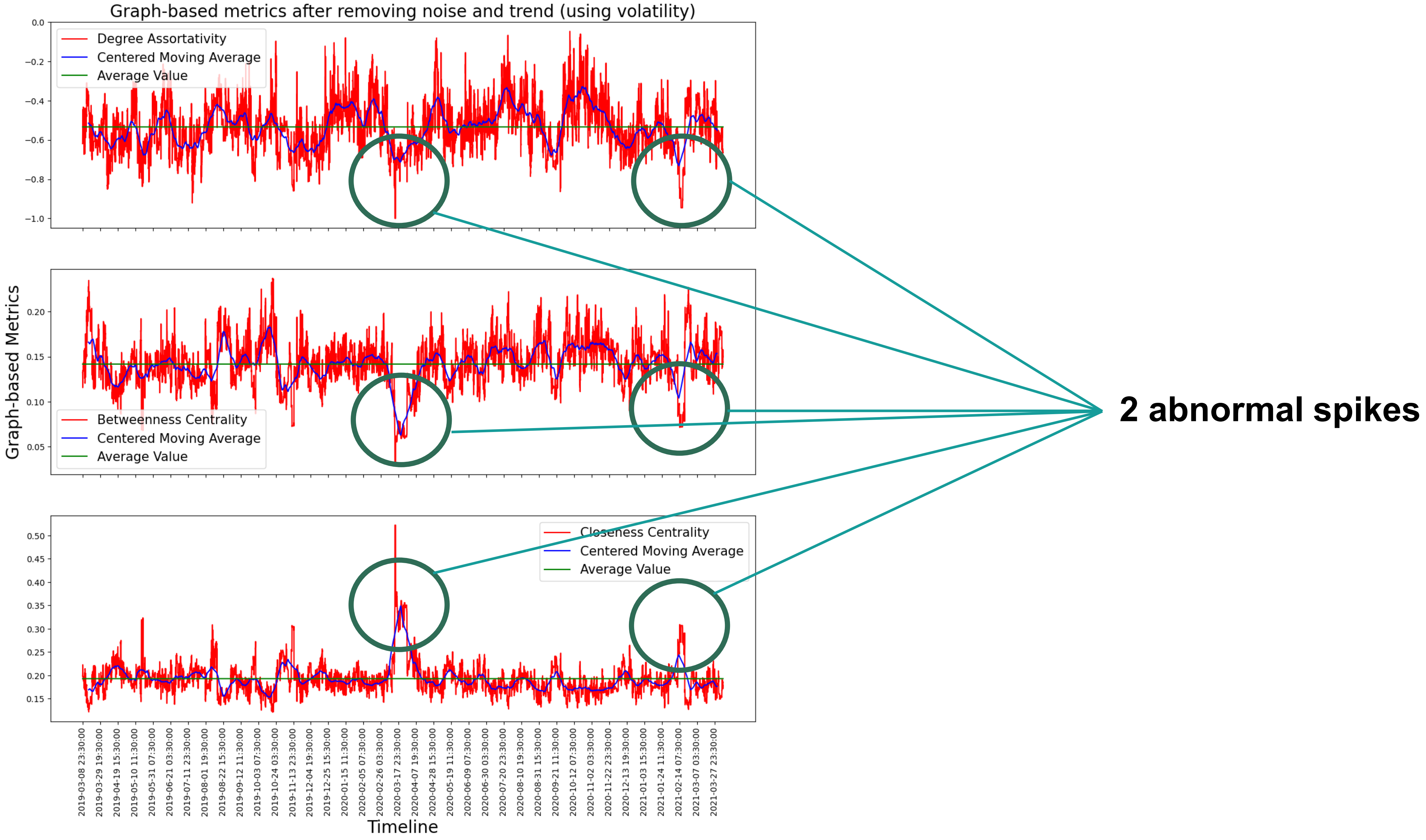
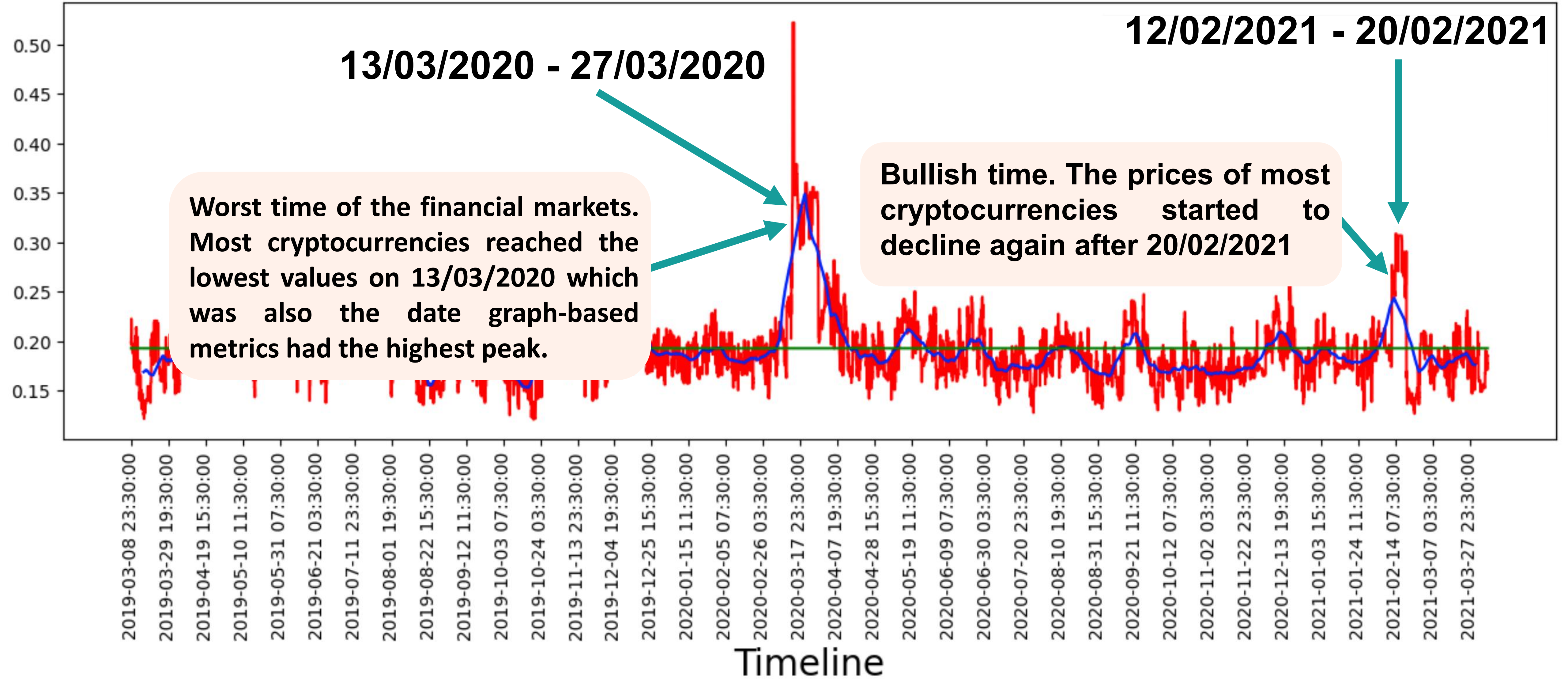


Community Structures in the Cryptocurrency Market

Results



Community Structures in the Cryptocurrency Market

Conclusion 2

The changes in a network's structure help identify abnormal events in the market.

Community Structures in Cryptocurrencies, Stocks and US ETFs

Target

How cryptocurrencies, stocks and US ETFs are correlated to each other?

More underlying characteristics can be discovered?

Community Structures in Cryptocurrencies, Stocks and US ETFs

Dataset

- 222 assets categorized into 3 types of investment vehicle:
  - Cryptocurrencies: 27 assets
  - Stocks: 146 assets
  - US ETFs: 49 assets

- Period is from 01/04/2019 to 03/05/2023

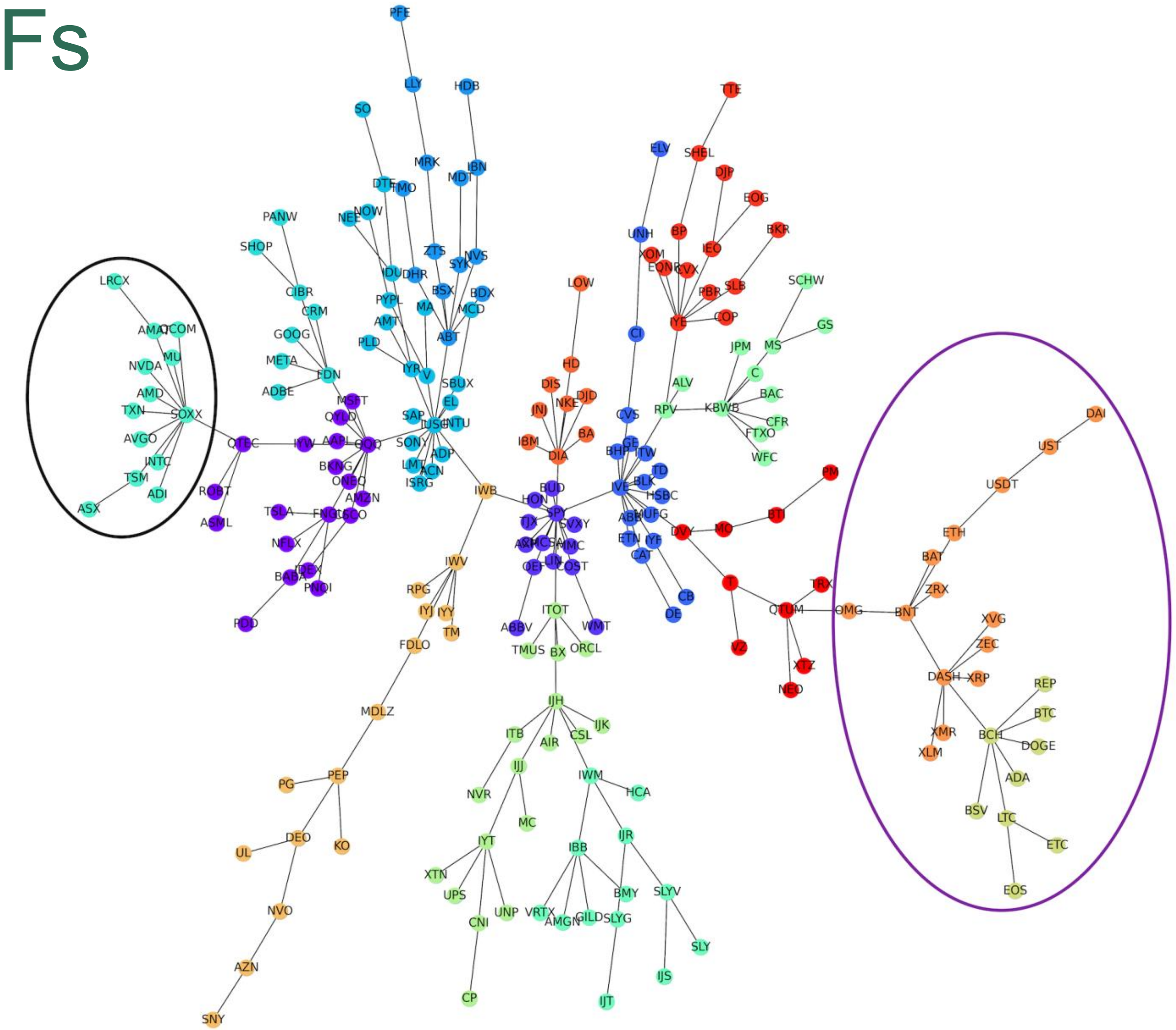
- Timescale: 30 mins

- Official trading time: 09:30 a.m to 16:00 p.m, Monday to Friday

- Timeline Division

Name	Pre-Covid-19	Covid-19 pandemic	Post-Covid-19	Bull market	Ukraine-Russia conflict
Time interval	01/04/2019 to 31/12/2019	01/01/2020 to 30/06/2020	01/07/2020 to 31/12/2020	01/01/2021 to 30/04/2022	01/05/2022 to 03/05/2023

Community Structures in Cryptocurrencies, Stocks and US ETFs



Pre-Covid-19
(04/2019 – 12/2019)

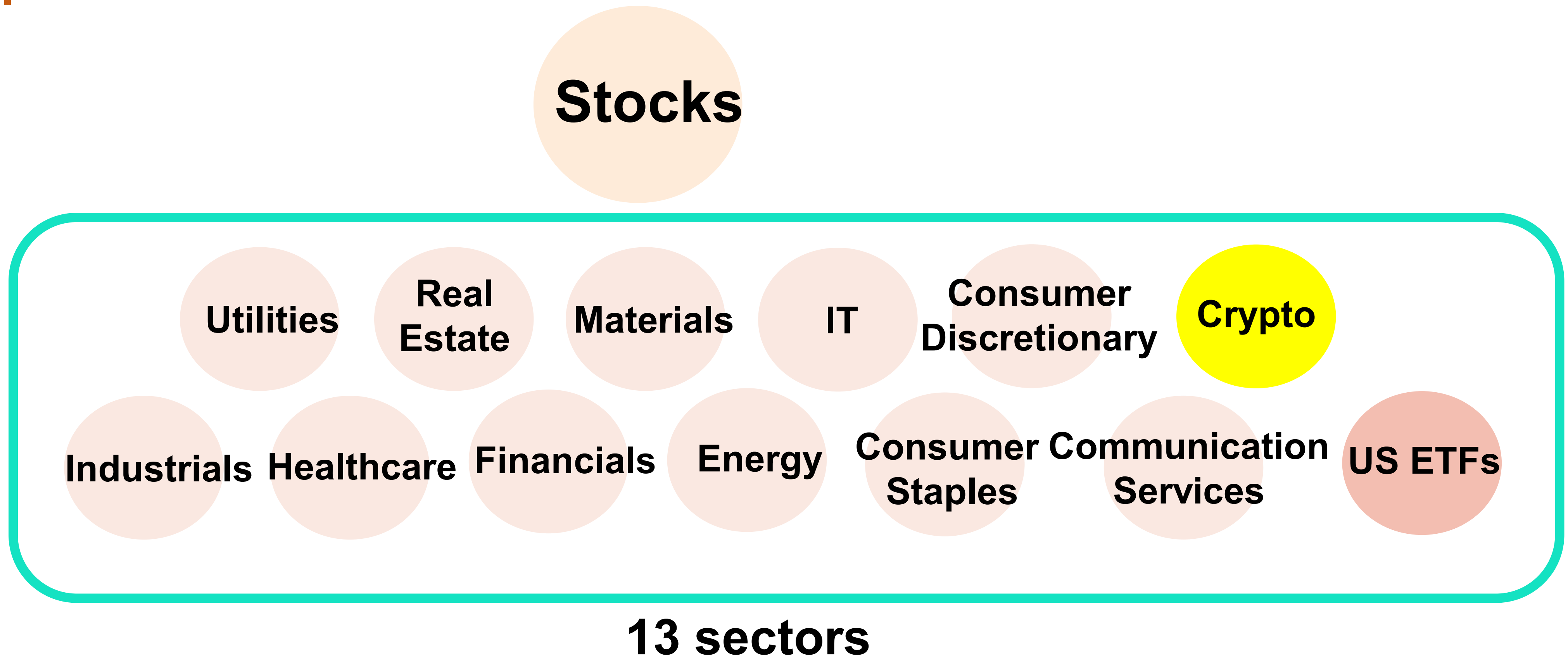
Results

US ETFs and stocks cluster together and US ETFs tend to be the central node of a community, linking other stocks

Cryptocurrencies reveal distinct behaviour from stocks and US ETFs

Community Structures in Cryptocurrencies, Stocks and US ETFs

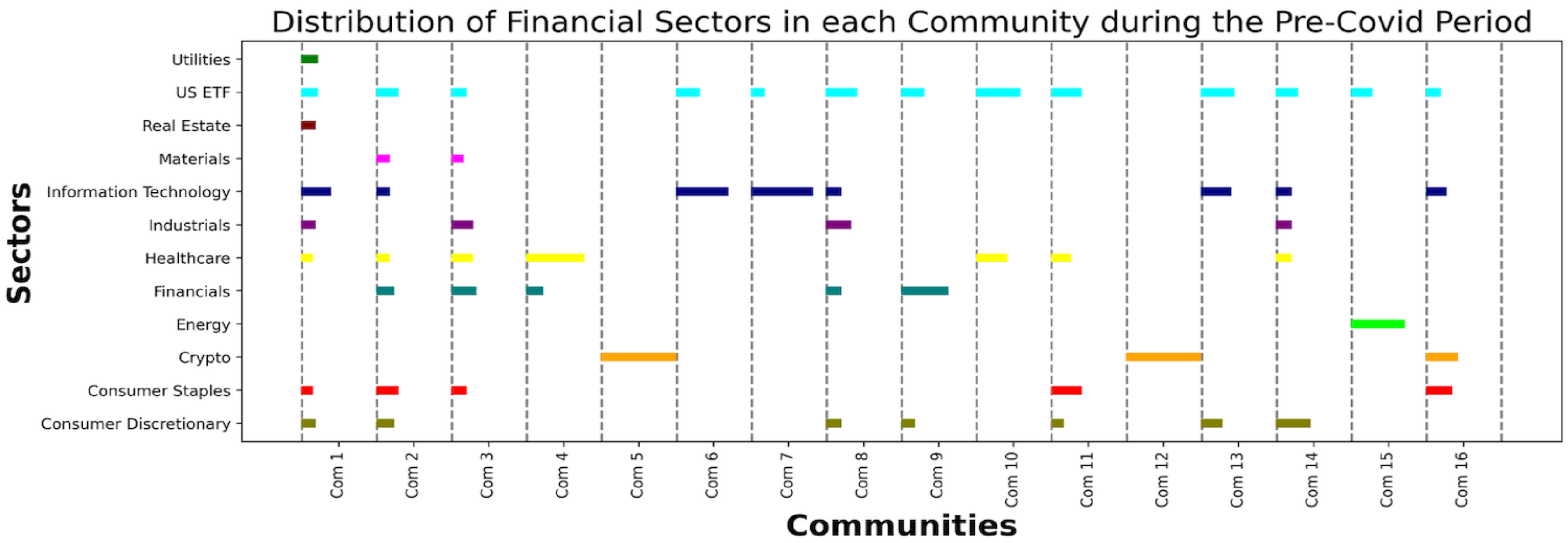
Sector Separation



Community Structures in Cryptocurrencies, Stocks and US ETFs

Sector Separation - Results

b)



Apart from cryptocurrencies, 4 sector sectors, namely technology, energy, financials and healthcare tend to form communities among themselves

Community Structures in Cryptocurrencies, Stocks and US ETFs

Conclusions

Cryptocurrencies show distinct characteristics from traditional assets, providing a potential of being safe-haven/hedge assets.

Some stock sectors, namely technology, energy, financials and healthcare are more sensitive to market conditions (e.g. downturn, bear/bull, market crash, etc).

From Community Structures to Investment Strategies